An error occurred on this slide. Check the terminal for more information.

The Vision

What if you could have a real conversation with Aristotle, Turing, or Marie Curie—and they actually knew their life's work?

🧠

150+ AI Faculty Live

Scaling to 10,000. LLM personas trained on complete corpora via RAG—books, letters, speeches

🎤

Speaking Busts

Real-time TTS + **Audio2Face** animation for lifelike dialogue

👥

Socratic Learning

Structured courses with verifiable micro-credentials (W3C Open Badges)

GPU-intensive workloads: Every conversation requires LLM inference, vector search, TTS synthesis, and

GPU Compute Architecture

Real-time AI pipeline requiring sustained GPU inference

LLM Inference

- • **NVIDIA NIM** for Llama/Mistral
- • **TensorRT-LLM** optimization
- • Streaming responses
- • ~500ms first token

`A100/H100 • 40-80GB`

RAG Pipeline

- • Vector embeddings
- • **pgvector** (CPU-based, Supabase)
- • 1,000+ Faculty corpora
- • ~50ms retrieval

`CPU (Supabase) • Optional GPU acceleration`

Conversational AI

- • **PersonaPlex** (NVIDIA, 2026)
- • Full-duplex: listen + speak
- • Voice + role per Faculty
- • Natural interruptions

`Alternative: Riva TTS • T4`

Audio2Face

- • **NVIDIA Omniverse**
- • Real-time lip sync
- • Emotion blending
- • 52 ARKit blendshapes

`RTX 3060+ / T4 • 8GB+`

NVIDIA Technology Stack

Current Stack (Pre-NVIDIA)

LLM OpenRouter API
Currently using hosted LLMs. → **Migrate to NIM**

TTS Edge TTS
Basic speech synthesis. → **Migrate to PersonaPlex or Riva**

AVT CSS Animation
Simple lip movement. → **Migrate to Audio2Face**

Target NVIDIA Stack

A2F Audio2Face + Omniverse
Real-time facial animation for Speaking Busts. **First Integration Target**

NIM NVIDIA NIM + TensorRT-LLM
Optimized Llama/Mistral inference for Faculty dialogue

PPX PersonaPlex (NVIDIA ADLR, 2026)
Full-duplex conversational AI with voice + role control. **Perfect for Faculty personas**

Riva NVIDIA Riva + Triton
Alternative: GPU-accelerated TTS with voice cloning per Faculty

Why NVIDIA? Real-time multi-modal AI (text + voice + avatar) requires unified GPU stack. **PersonaPlex** (2026) or **Riva** + custom pipeline: full-duplex conversations with custom voices + roles—perfect for 10,000 Faculty personas. Today: <500ms LLM + <100ms

What We've Built

Platform MVP complete • Ready for GPU scaling

150+
AI Faculty Live
Scaling to 10,000 via RAG

1:1
SAMWISE AI Tutor
Per student, memory-bearing

3D
Speaking Busts
CSS animation → Audio2Face

7+
Learning Experiences
Symposia, Salons, Debate, etc.

Frontend

Next.js 15, TypeScript, Tailwind, Three.js (for busts)

Backend

Supabase (Postgres, Auth, Edge Functions), pgvector

AI Pipeline

OpenRouter → target NVIDIA NIM, Edge TTS → target Riva

Current blocker: Running on API providers. Need dedicated GPU compute for production scale + Audio2Face integration.

Learning Experiences & Publishing

7+ structured learning formats powered by Faculty dialogue

Encyclopædia

13 volumes, 483+ entries – Adult & Children's editions with interactive flipbook reader

Symposia

7-session guided dialogues

Salons

Intimate conversations

Debate

Structured disputation

Drama

Embodied role-play

The Inquirer

Quarterly journal – 2 volumes, 4 issues, 20+ articles in press

iNQ Publishing

Illustrated classics – Wm. Blake Commedia, Mucha-style editions

Platforms & Ecosystem

Atlas (prototype)

atlas.castalia.institute – Curriculum & learning maps, standards alignment

Terrain (prototype)

terrain.castalia.institute – Practice & assessment terrain

iNQspace (open source)

inqspace.castalia.institute – GitHub Classroom-style delivery; **auto-grading via GPU inference**

AIMA / AIPA

Structured certificates with W3C Open Badges. Verifiable micro-credentials.

iNQ Cards

Physical inquiry cards (line art + QR back); catalog live. Integrates with Ægis.

Together: **explore** (Atlas) → **practice** (Terrain) → **submit & grade** (iNQspace) → **certify** (AIMA/AIPA)

Atlas, Terrain & iNQspace

Atlas (prototype)

atlas.castalia.institute

- • Curriculum & learning maps
- • Standards alignment (K–12, institutional)
- • Pathways to Faculty & courses
- • Navigate the landscape of knowledge

Terrain (prototype)

terrain.castalia.institute

- • Practice & assessment
- • Structured scenarios, mastery
- • Progress through AIMA/AIPA
- • Verifiable credentials (Open Badges)

iNQspace (open source)

inqspace.castalia.institute

- • GitHub Classroom-style course delivery
- • PR-based submissions & inline grading
- • **Auto-grading via GPU inference**
- • Devcontainer support, fork sync

Together: **explore** (Atlas) → **practice** (Terrain) → **submit & grade** (iNQspace) → **certify** (AIMA/AIPA)

Micro-Credentials & Virtual Campus

AI-native credentialing with GPU-accelerated tutoring, grading, and an immersive exploration environment

Castalia Micro-Credential System

Each micro-credential (10–25 hrs) integrates **SAMWISE** AI tutoring, **Beatrice** AI grading, GitHub assignments, and portfolio artifacts.

Academic ladder: **Apprentice → Adept → Magister → Fellow → Custodian**

12 core AI credentials + specialization + capstone = Certified Magister

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GPU needs: Beatrice grading runs LLM inference for code evaluation, artifact verification, and credential issuance at scale.

Map of Inquiry & WorkAdventure Campus

Curriculum as a navigable knowledge map: **cities** = learning activities, **regions** = disciplines, **paths** = prerequisite relationships.

Explored via **WorkAdventure** – an immersive 2D virtual campus where students walk the Map, encounter Faculty bots powered by NIM, and collaborate spatially.

GPU needs: Faculty bots in WorkAdventure require real-time LLM inference + TTS for spatial conversations as students explore.

PersonAI Integration

Specializations

AI Faculty Network

iNQ Cards – Physical-to-Digital Bridge

Physical inquiry cards with a tiered artifact model. Every card resolves to a single URL – **QR scan, NFC tap, or typed URL all reach the same GPU-powered knowledge layer.**

Product Stack

- • **QR Cards** – card stock, matte finish, $20/mo subscription packs
- • **NFC Edition** – embedded NFC tag, tap to open; collector decks
- • **Bamboo Artifact** – laser-engraved, optional NFC inlay; $40–80/deck
- • **Metal Magister** – etched brass/bronze micro-credential medals

GPU-Powered Backend

- • **Faculty Cards** – scan Darwin, Hypatia, Euler → unlocks AI conversation with that persona via LLM inference
- • **Expedition Cards** – Yellowstone, Amazon, Moon → maps & missions on Atlas
- • **Ægis integration** – camera sees card back → ambient response via on-device + cloud inference

inquiry.institute/card/photosynthesis – same digital knowledge layer powers all physical formats. Manufacturing phases: QR print → NFC packs → bamboo & metal artifacts.

SAMWISE: AI Learning Companion

Supportive Assistant for Memory, Writing, Inquiry, Sense-making, and Exploration

What SAMWISE Is

Personated

Consistent voice and presence—a named companion, not an anonymous tool

Inquiry-First

Asks thoughtful questions rather than providing ready answers

Adaptive

Adjusts to each learner's developmental stage

Memory-Bearing

GPU Compute Needs

- • **LLM Inference** – Continuous dialogue with context retention
- • **RAG Memory** – Vector search across learner's conversation history
- • **PersonaPlex** – Full-duplex conversations with natural interruptions
- • **Long Context** – Multi-session memory spanning weeks/months

"I can't carry it for you, but I can carry you."

– Samwise Gamgee

Market Opportunity

$350B

TAM

Global online education

(HolonIQ 2025)

$20B

SAM

AI tutoring + adult learning

(Market research)

$100M

SOM

Premium learners (engineering, science, music, humanities)

(5-year target)

Why Now?

Team

Daniel McShan, PhD

Founder / Custodian

- **PhD Computational Biosciences** (CU Med)
- **25+ years NASA**: Orion, Mars Phoenix, Stardust
- CTO Formul8.ai (AI/ML), VP Eng AVS (aerospace)
- 5 publications, 1 patent

AI Development Partners

- **Claude** (Anthropic) – Architecture
- **Cursor IDE** – AI-first development
- **GPT-4** – Content generation

Actively Recruiting

- **ML Engineer** – LLM optimization, NVIDIA stack
- **Content Lead** – PhD (engineering, science, music, or humanities), curriculum
- **Advisors** – EdTech, AI ethics, NVIDIA ecosystem

AI/ML | Real-Time Systems | Full Stack | GPU Compute

1,000+ AI Faculty Across 10 Colleges

Form & Number	Matter & Life	Value & Virtue	Mind & Machine	Word & Meaning	Learning & Becoming	Time & Place	Power & Change	Order & Chaos	Beauty & Form
a.euclid	a.darwin	a.aristotle	a.turing	a.shakespeare	a.montessori	a.herodotus	a.smith	a.popper	a.michelangelo
a.newton	a.mendel	a.plato	a.vonneumann	a.homer	a.dewey	a.thucydides	a.marx	a.kuhn	a.davinci
a.einstein	a.pasteur	a.kant	a.shannon	a.dante	a.pestalozzi	a.gibbon	a.keynes	a.prigogine	a.picasso
a.gauss	a.curie	a.nietzsche	a.mccarthy	a.virgil	a.froebel	a.tacitus	a.hayek	a.mandelbrot	a.monet
a.leibniz	a.watson	a.confucius	a.minsky	a.chaucer	a.freire	a.machiavelli	a.weber	a.lorenz	a.vanGogh
a.pythagoras	a.crick	a.buddha	a.newell	a.dickens	a.vygotsky	a.toynbee	a.durkheim	a.wiener	a.bach
a.archimedes	a.franklin	a.aquinas	a.hopper	a.tolstoy	a.papert	a.braudel	a.mill	a.bateson	a.mozart
a.galileo	a.linnaeus	a.descartes	a.lovelace	a.joyce	a.bruner	a.rank	a.rousseau	a.maturana	a.beethoven
a.planck	a.hooke	a.hume	a.babbage	a.woolf	a.piaget	a.carlyle	a.locke	a.varela	a.stravinsky
a.heisenberg	a.leeuwenhoek	a.russell	a.dijkstra	a.borges	a.skinner	a.spengler	a.hobbes	a.holland	a.cage
... +90 more	... +90 more	... +90 more	... +90 more	... +90 more	... +90 more	... +90 more	... +90 more	... +90 more	... +90 more

150+ AI Faculty in production today – scaling to 10,000 (1,000 per college × 10 colleges) with complete RAG corpora, LoRA fine-tuning, and Audio2Face integration

GPU Scaling Roadmap

Phase 1: Integrate

With Inception → 3 months

- Migrate LLM to **NIM**
- Integrate **Audio2Face**
- **50 users** private beta
- 1× A100 instance

~$2K/mo
GPU compute

Phase 2: Launch

Months 3–9

- **500 users** public beta
- 2× A100 + load balancing
- **PersonaPlex** or **Riva** TTS
- Full Audio2Face pipeline

~$8K/mo
GPU compute

Phase 3: Scale

Months 9–18

- **5,000+ users**
- **DGX Cloud** cluster
- **Triton** inference server
- Multi-region deployment

~$25K/mo
GPU compute

How Inception Credits Help

- Offset compute costs during critical 0→1 phase
- Access to NVIDIA engineering support
- Early access to PersonaPlex, NIM + Riva optimizations

Unit Economics at Scale

- Revenue: $35 ARPU/user/month
- Compute: $5–10/user/month (optimized, pooled sessions)
- **70% gross margin target**

Inception Member – Next Steps

How We'll Leverage Inception

GPU Credits

$25K-50K

DGX Cloud or partner credits for Phase 1-2 (12 months)

Technical Support

Audio2Face + NIM

Guidance on integration, optimization, best practices

Ecosystem Access

Partners + GTC

Co-marketing, case study, potential GTC showcase

We Commit To

- • Showcase NVIDIA tech in production (case study)

12-Month Goals

- • 1,500 paying users on NVIDIA-powered infrastructure

Let’s Build Together

GPU-accelerated conversational AI for engineering, science, music, and liberal arts education

Email
custodian@castalia.institute

Website
castalia.institute
atlas · terrain · inqspace · aima · aipa

Demo
castalia.institute/demo

Castalia Institute is an **NVIDIA Inception** member, with **AWS Activate** and **Google Cloud for Startups** backing our cloud infrastructure—accelerating our mission of making the world's greatest thinkers accessible to everyone.